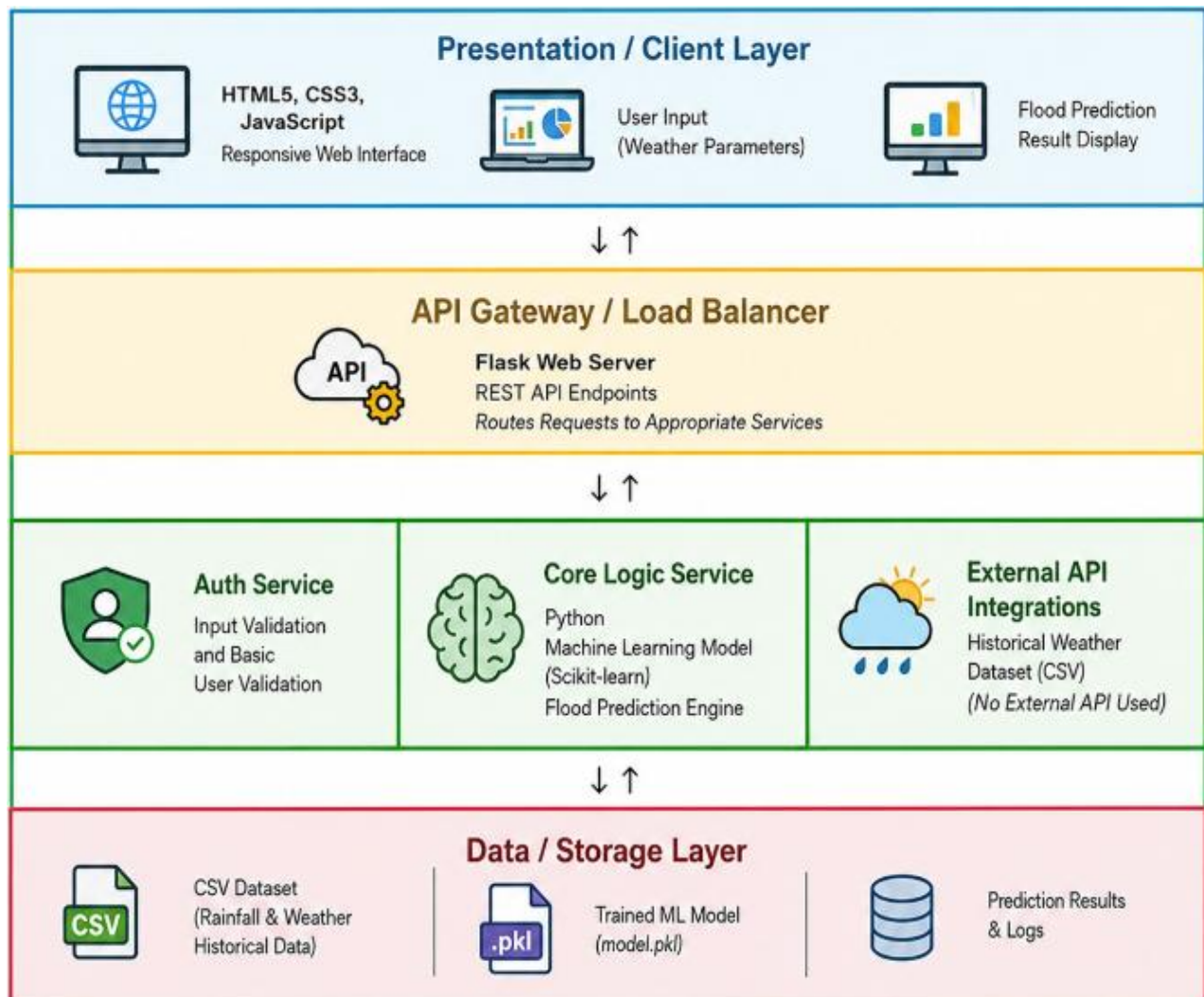


Solution Architecture

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for entering weather parameters and displaying flood prediction results</i>	<i>HTML, CSS, JavaScript</i>
<i>API Gateway</i>	<i>Receives user requests and forwards them to the machine learning prediction service.</i>	<i>Flask</i>
<i>Microservices</i>	<i>Processes weather data, executes the ML model, and generates flood prediction results</i>	<i>Python, Scikit-learn</i>
<i>Database</i>	<i>Stores historical rainfall/weather dataset and trained machine learning model.</i>	<i>CSV Dataset, Pickle(.pkl)</i>